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Fragility analysis of structures subjected to seismic excitation: A state-of-the-art review

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ABSTRACT

Recent concerns have witnessed that excitation due to a seismic event has caused catastrophic destruction to mankind and its existing surroundings. Loss of livelihood and substantial impact on economic loss in various parts of the world became a primary concern. The occurrence of damage in structures is one of the main reasons to study their seismic behavior when subjected to seismic excitation. Fragility analysis (FA) of structures provides a better understanding of the response of the structural and non-structural systems during seismic events. When subjected to the varying intensities of ground shaking, the probability of exceeding a particular damage state is defined by the fragility curves. This paper is focused on the in-depth overview of the studies conducted in the domain of fragility analysis of structures by several researchers. The adopted analysis methods in past studies include non-linear static pushover analysis, dynamic time history analysis, incremental dynamic analysis, etc. The review study comprises of fragility analysis of structures categorized into different classes, precast structures, with or without the influence of soil-structure interaction, buildings located in the region of hills, transmission towers, concrete face rockfill dams, etc. The outcome of this study is to benefit the research community provide a collective idea of this domain and identifying the least concerned area that could be given more priority in the near future.

1. Introduction

Damage caused by an earthquake (EQ) is catastrophic. The sudden seismic event eventually leads to disturbance to mankind and its surroundings. The damage of an EQ begins with a vigorous shaking of the earth. Tremors of a high-intensity EQ can be felt hundreds of kilometers away. Shaking of ground or ground movement is the primary cause of EQ damage to man-made structures. Factors like EQ intensity, the location's proximity to damage, local geological conditions, and the type of soil greatly influence the extent of seismic excitation.

EQ vibrations can cause the earth to shift and tear its surface. Disturbed surfaces can cause other hazards including damage to roads and existing structures. EQ of different strengths affects structures to various degrees. Natural hazards can never be avoided, but damages caused by natural hazards can be prevented to a certain extent if preventive measures are taken. Similarly, if the probable damages that may occur against a certain magnitude of EQ can be predicted, thousands of lives can be saved by preventing the collapse of structures. Ductility plays an essential role in determining a structure's seismic performance and allowing it to withstand shocks and vibrations. EQ resistant design aims to provide reasonable ductility to the structure by appropriate ductile detailing. Therefore, attempts should be made to design EQ-resistant structures instead of EQ-proof structures. Because the former is more economical than the latter and ensures safety against disasters.

An EQ design philosophy is illustrated in Fig. 1. With slight and frequent shocks, the main elements of the building carrying the vertical and horizontal forces must not be damaged. However, components that do not hold the load can experience irreparable damage. Major elements can be damaged irreparably in moderate but occasional shaking, requiring replacement after an EQ. Strong but less persistent shocks can cause severe (even irreparable) damage to major structural elements, but the building must not collapse. Therefore, after slight shocks, the building should be fully functional in a short time, and repair costs should be low. In moderate shaking, after completion of repair and

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replacement of the damaged structural elements, the building could be put into operation. Whereas, post-high-intensity EQ, the structures may become non-functional for further utilization, yet they shouldn't be collapsed so that people can be evacuated and property can be restored [1].

1.1. Recorded major structural damages in the past earthquakes

An EQ is a naturally occurring event due to tectonic plates' movement, i.e., when there is a sudden slip on the fault. This movement further causes massive intrusion on mankind. Loss of livelihood and enormous economic impact have been witnessed with recorded EQs. Table 1 shows various structural damages caused by EQs in the past decades.

Certain extensive seismic events have been recorded, which caused catastrophic destruction among the existing life on earth. For example, an EQ with a magnitude of 9.5 hit Valdivia, Chile, in 1960. It was the most extreme seismic event ever recorded in the history of mankind. Due to the high magnitude, it disturbed the overlying ocean, which generated tsunamis. The aftershocks and after effects created havoc with huge loss of lives, including substantial damages. The damages largely influence the casualties and mortality rate that occurred in the structure during a particular seismic event. So, the structure must withstand any natural hazard to protect the occupants residing in them. Therefore, EQ-proof or resisting structures can reduce damages and protect the occupants. Thus, the functionality of a structure depends on the extent of damage induced by a particular EQ.

Various structural damages observed during a seismic event includes the following [3]:

- The collapse of brick walls of the reinforced concrete (RC) structures while the infrastructures are held up.
- Failure of slab roof was proven deadly in case of heavy clay roofing.
- Traditional houses made up of mud bricks were witnessed with total collapse.
- Recently constructed engineered buildings were found to be withstanding the consequences of the seismic event in contrast to non-engineered buildings that were found to be crumbled.

1.2. Introduction to fragility analysis

To recognize the responses of a structure when subjected to varying intensities of ground motion caused due to an EQ, seismic fragility analysis (FA) plays a significant role. It can be defined that the inter-story drift ratio (maximum lateral displacement between two successive floors, normalized to the floor height), global drift ratio (maximum roof deviation normalized to the height of a building), or story shear force, etc., as damage limit states in fragilities.

In recent times among the researchers, fragility analysis (FA) has been the spotlight for study on seismic behavior of different structures.

Table 1
Structural damages caused by EQ in past decades [3].

Month	Year	Local Magnitude in Richter's	Location	Structural Damages
Dec	2003	6.6	Bam, Iran	Masonry, clay bricks
Oct	2005	7.6	Kashmir, Pakistan	Slab roof, concrete
Aug	2007	7.9	Ica, Peru	Brick and concrete
May	2008	8.0	Sichuan, China	Concrete and brick
Jan	2010	7.2	Port-au-Prince, Haiti	Blocks of cement Concrete, stacked bricks

The application of FA helps in deriving fragility curves (FC), which aid to provide a better understanding of the structure's behavior corresponding to a particular seismic event and estimating the seismic fragility of a structure. Fig. 2 represents a simplified FC, where the P_f (probability of exceedance) elevates with an increase in peak ground acceleration (PGA) [4].

To assess the seismic losses and control the city risk, seismic fragility analysis of structures plays a key role. This analysis type helps to understand the response of a structure due to EQs and determine the damages of a structure correlating to a particular intensity level of EQ. Therefore, fragility analysis methods have been emerged as a vital reference to mitigate the impact of an EQ on structures.

The fragility curve can be determined using the following methods:

- Professional expert judgment
- Quasi-static analysis and code consistent analysis
- Statistical analysis of damage data related to past EQs
- Numerical simulation based on structural dynamics analysis

Because of inadequate & poor reliability of data after EQs, numerical methods are the most widely accepted methods in existing studies or research.

Several researchers have proposed guidelines and methodologies in

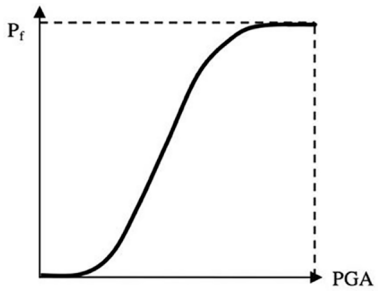


Fig. 2. Schematic of a seismic fragility curve [4].

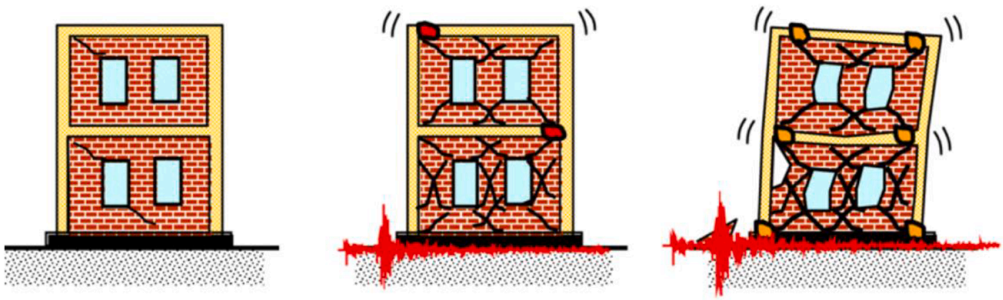


Fig. 1. (a), (b), and (c) shows the performance objectives under EQ shaking of different intensities [2].

recent years to assess the vulnerability of building structures through fragility analysis [5,6,7]. This has become a necessary tool to study the seismic loss excitation of the building structure. In addition, seismic design codes provide the design criteria that must be followed to design seismic-resistant structures.

When comparatively analyzed, seismic standards such as Brazilian Standards, American Standards, Eurocode 8, Italian Code, etc., revealed that the required significant properties of a seismic-resistant structure include symmetry, homogeneity, simplicity, redundancy, etc. However, the detailing and design of a structure should primarily be focused on contributing adequate ductility, as it plays a significant role in the energy dissipation within a structure in the non-linear range [8].

1.3. Damage states

Damage states (DSs) for a building's structural and non-structural systems are described separately. The structural damage is distinctively classified as "Slight, Moderate, Extensive, or Complete and Collapse", with collapse being the subset of complete structure damage according to EQ's HAZUS loss estimation. Similarly, the four damage states were separately defined for the non-structural components. Since a continuous scale is impractical, damage states characterize damages of building structures. In addition, it provides an idea of the physical condition of a building to the consumer. Loss functions link the physical state of a building to various loss parameters (i.e., direct economic loss, casualties, and loss of function) [5].

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Table 2 illustrates the four structural damage states and some data from past studies. Performance-based material's damage index is represented in [9], whereas the damage indices of limit state are shown in [10]. Meanwhile, standard deviation (β) parameters are provided in [11], and the mean (%) of story drift ratio capacity is represented in [12]. Several researchers adopted double parameter damage indices by Park & Ang to evaluate the damages.

2. Fragility analysis of structures: Scope of previous research

Recent trends have witnessed the rising interest in the structure's behavior under the effect of EQ occurrences. This paper presents a brief literature review in the domain of FA of structures. Several researchers proposed methodologies and studied the performance of different structures considering varying conditions to depict the nature of the structure's outcome.

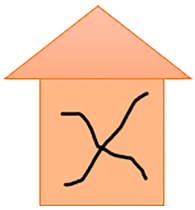
The stochastic method developed by Lin [12] generates a simple solution for the mean value and standard deviation of the peak story drift of a frame structure for linear elastic structures. Whereas, to account for each story's response during seismic excitation, story-specific demand models for multi-story buildings were developed by Bai et al. [13]. For high-intensity EQ, fragility estimation based on the traditional demand model was lower than those based on the proposed story-specific demand models. For lower intensity EQ, they can be more significant. The methodology derived reveals that the selected buildings (i.e., 2 and 5 stories) are incredibly vulnerable to ground vibrations, with a 2% chance of exceeding in the next 50 years. For both structures, the FC derived using story-specific demand models had steeper slopes than those created utilizing traditional demand models.

A large-scale vulnerability of seismic evaluation of RC buildings using an analytical technique was carried out where it allows the use of inter-story drift as a demand parameter of structure for the computation of damage. As a result, FC were generated, allowing the annual failure probability to be calculated at the specified DSs [14].

On the contrary, Macabuag et al. [15] suggested that existing fragility evaluations for the absolute analysis of subsets of data (such as building material), data that are aggregated or ordinary least square (OLS) parameter estimates be re-examined for potential bias. The particular finding has significant implications for the succeeding production of analytical fragility functions (FF) and empirical, as it introduces numerous concepts new to the field of assessment of fragility. Additionally, it finds the force measures, including the regime of low, giving the most efficient Tsunami Intensity Measure (TIM) for high precision inundation simulations.

Fig. 3 shows the number of papers published in the last two decades from the database considered for this review. It can be seen that the research trend in FA of building structures started elevating in the year 2015, with the greatest number of papers being published in the year 2020. Further, it dropped in the year 2021, which depicts that there is still further scope in the domain of this study. Additionally, Fig. 4 summarises the adopted analysis types incorporated in this review

Table 2
Structural damage States.

Damage States	Slight	Moderate	Extensive	Collapse
Illustration				
Description	Non-structural hairline (diagonal or horizontal) cracks in infill walls; maybe cracks at the frame of an interface of infill wall	Severe damage of non-structural walls; extensive cracking in walls; maybe diagonal shear cracks in beams or columns	Partial or complete failure of the structural component; out-of-plane bulging of infill walls; maybe permanent lateral deformation	Partial or total failure/collapse of the structure; total failure of infilled walls; non-ductile failure of columns and beams of concrete.
[9]	0.15–0.3	0.3–0.6	0.60–0.90	>0.90
	≤ 0.4	0.4–0.7	>0.7	–
[10]	1/450	1/300	1/150	1/50
MIDR				
Park & Ang	0.1	0.25	0.4	1.0
Niu	0.2	0.4	0.6	0.9
Lu-Wang	0.2	0.4	0.6	0.9
[11]	0.18	0.24	0.31	0.15
[Mean %]	0.2	0.5	1.5	2.5
[12]				

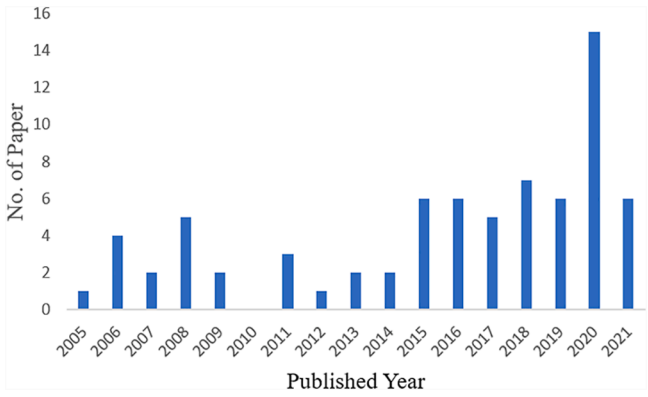


Fig. 3. Published papers on fragility analysis of structures during years 2005–2021.

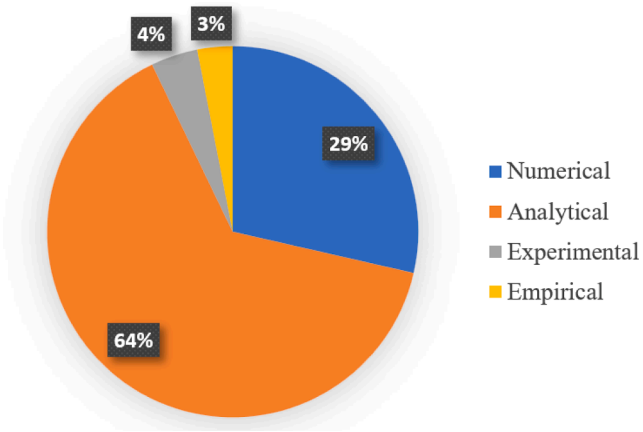


Fig. 4. Adopted types of analysis in the referred literature.

study.

2.1. Prerequisite of the study

The expanding impact of EQ that contributed to the considerable loss of economic, social impact, and livelihood has centered the attention to primarily focus study the Structural EQ behavior to help lessen the loss. Seismic fragility analysis has been in the spotlight in the recent determination of the structure's performance when it is subjected to varying levels of ground motion records that can be either from the past EQ records or can be generated synthetically. It is an effective tool for assessing the structural system. Adopting various static and dynamic analysis methods, the analysis of the building structure is carried out. The derived FC better understand the structure's response and probable damages. Therefore, this assessment depicts a key role in mitigating risk and provides an awareness of post-EQ planning. Fig. 5 shows the overview of different structures and their conditions incorporated in this study which is discussed in the later part of this review.

2.2. Adopted methods of analysis for the derivation of fragility curves

FCs are integral in determining the probability of exceeding the damage limit state or failure corresponding to the ground motion intensity measure. The vulnerability of seismic can be determined based on the derived FCs. The different adopted analysis methods are discussed in detail in the later content of this study.

Fig. 6 sums up the adopted method of analysis throughout the entire content of this review study. According to Fig. 6, incremental dynamic analysis (IDA) could be observed as the most adopted method of analysis in this review study.

2.2.1. Non-linear static pushover analysis

Pushover analysis (PA), also known as non-linear static analysis, is a method for evaluating structures' inelastic strength and deformation and exposing structural design flaws. Its key benefit is that it makes it easier for structural engineers to identify crucial seismic response values and use engineering considerations to precisely adjust the power, capacity, and deformation quantities that govern seismic response near damage. It should be noted that this analysis is approximate and does

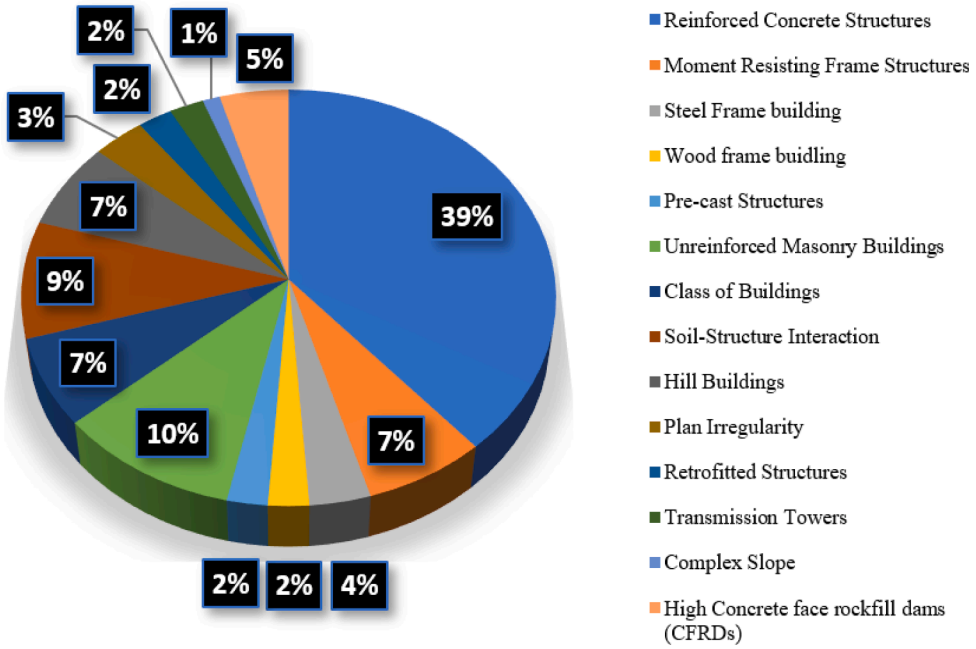


Fig. 5. Adopted structural typologies and their conditions in the referred literature.

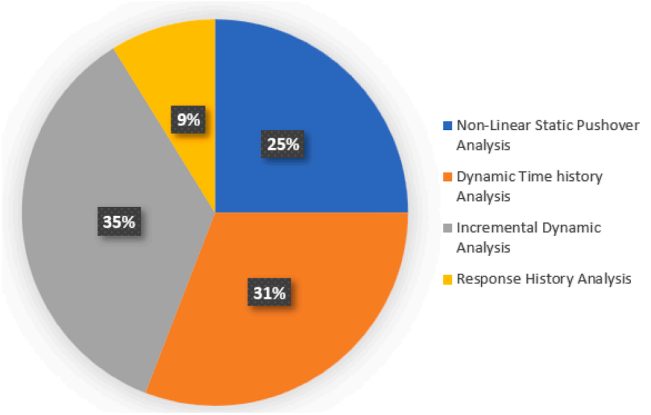


Fig. 6. Adopted method of analysis in the referred literature.

not account for complex properties such as hysteresis, higher participation mode, and so on. Nevertheless, it's well-known for producing good results in typical structures (without torsional irregularity). The junction of the building capacity and demand spectrum curve shown in Fig. 7 (i.e., the performance point) determines the structure's response [16]. The prime target is the precise prediction of the peak response of buildings and components for the considered seismic event.

This method is well known for its simplicity while performing the analysis. The main reason behind considering PA is that it provides a more excellent grasp of the behavior of the building. Its main contribution includes identifying weak elements and realistic prediction of demand models. Many researchers considered PA due to the criteria mentioned above.

According to the building inventory, the vulnerability analysis was performed in the district of Arenella in Naples, where major residents were of moment resisting frame of RC structures. The fragilities considered the uncertainties of the surveyed parameters and the heterogeneity of the limit-state thresholds and inelastic demand. It was found that geometrical configuration dispersion and material properties have a relatively minor impact on vulnerability for all groups of buildings studied [17]. However, Lagaros [18] identified how effective fragility analysis was at assessing the multi-story RC building's performance of seismic built to modern codes. He considered two sets of examples; one set including FI, weak ground storey, short-column. The other set consists of six bare-frame RC building cases with behavioral factor q, which ranges from 1.0 to 6.0 with a difference of 1.0 each. The authors mentioned above derived the fragility curves to evaluate the probability of exceedance for four damage states: slight, moderate, extensive, and collapse. As in [5], the FC is function of lognormal in which the probability of exceeding a particular damage state (d_s) given any seismic demand parameter or considering spectral displacement (S_d), is defined by Eq.1.

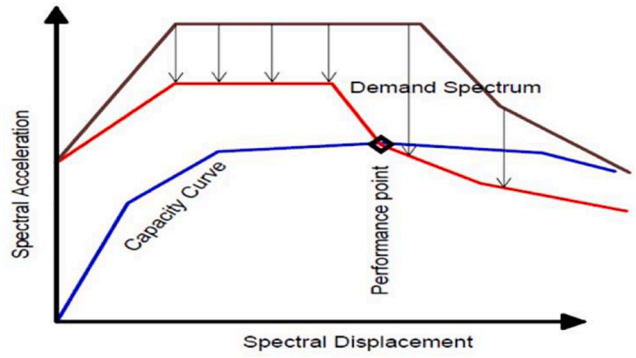


Fig. 7. Demand spectrum, capacity curve, and performance point [16].

$$P[d_s|S_d] = \Phi\left[\frac{1}{\beta_{ds}}\ln\left(\frac{S_d}{\overline{S}_{d,ds}}\right)\right] \quad (1)$$

Where $\overline{S}_{d,ds}$ is the median value of spectral displacement at which the building reaches the threshold of d_s; β_{ds} is the standard deviation of the natural logarithm of spectral displacement for d_s and Φ is the standard normal cumulative distribution function as indicated in Eq. (1).

$$\beta_{ds} = \sqrt{(\mathrm{CONV}[\beta_c,\beta_D])^2 + (\beta_{T,ds})^2} \quad (2)$$

Here, β_c shows the lognormal standard deviation parameter that expresses the capacity curve's total variability; β_D represents the lognormal standard deviation parameter that describes the demand spectrum variability, and $\beta_{T,ds}$ is the lognormal standard deviation parameter that describes the variability of a threshold of d_s as indicated in Eq.2.

Drift-based FCs were derived for confined masonry (CM) walls, the primary structural components of lateral load-resisting systems used in Latin-American housings. The derived FCs could be incorporated to determine limiting drifts associated with damage-based limit states for CM buildings, where the influence of brick type, horizontal steel reinforcement, and vertical compressive stress are considered. Fig. 8 represents FCs of CM walls without incorporating horizontal constructed wall with clay bricks unit, handmade of steel reinforcement. It shows that the influence of additional uncertainties such as aspect ratio of height to length and brick's mechanical properties that depend on specimen plays an essential role. Also, it was discovered that the limiting drifts stipulated in the Mexico City Building Code 2004 Edition for providing safety against collapse should be modified to include the types of brick [19].

Meanwhile, Fragility via Capacity Spectrum Assessment (FRACAS) is a new approach for deriving FCs. Compared to NLTHA, FRACAS appears to accurately capture both case-study structures' reactions. It was shown that FRACAS effectively reflects the impacts of record-to-record variability in FCs. It is simpler and faster than existing methods that directly employ accelerograms [20]. Furthermore, incorporating the same analysis method, Ruggieri et al. [21] revealed that the FCs estimated with flexible floors have a more significant probability of surpassing a specified engineering demand parameter (EDP) for a given IM. This implies that stiff floor modeling can produce non-conservative capacity-demand comparisons, particularly in retrofitted scenarios. At the same time, variability functions and FCs were established for RC buildings of the low, mid, and high rise in Patna, India, which is a seismically highly active location using PA. Multiple parameters with different combinations were considered, and modeling & analysis of the buildings were carried out using NLSPA. Infill frame (IF) structure variability was slightly higher than bare frame (BF) buildings. However, the ultimate spectral displacement of open ground storeyed (OGS) buildings was found to be lower than that of BF and IF buildings. It was discovered that regular buildings (square and rectangular-shaped) outperformed irregular (L-shaped) ones when evaluating the performance of structures based on various shapes [22].

However, Chandra Dutta et al. [23] developed FCs that reveal the lack of masonry infill on the ground floor level and poor design and construction quality, increasing sensitivity to the manifold. The findings could be beneficial in design offices. The methodology used could be adapted to diverse places for vulnerability assessment, utilizing a methodology that maintains a good balance between rigor and accuracy. The probabilistic study reveals that design construction quality (DCQ) does not affect the conditional likelihood of light damage state (DS1). At a low level of seismic hazard due to the increased strength and stiffness of fully infilled (FI) buildings, infill walls are useful even in structures with Poor design construction quality. At a high level of seismic hazard, partially infill (PI) framed buildings witnessed peak fragility, with 54 percent and 47 percent probability of complete damage, i.e., at PGA level of 0.36 g for 3 and 5-story buildings with good DCQ. This could

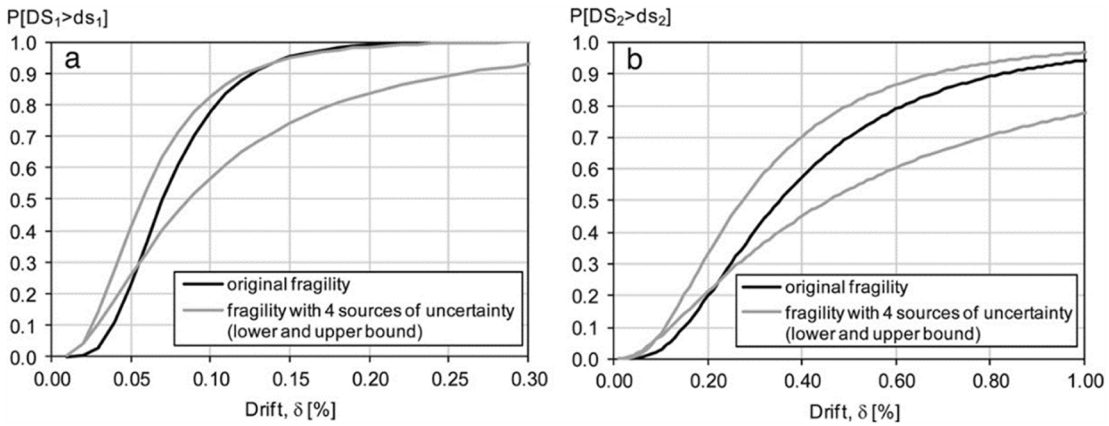


Fig. 8. Drift-based fragility curves for diagonal cracking (DS1) and complete X-shaped cracking (DS2) for CM walls [19].

help improve the design methodology for RC FI, PI and bare-framed structures.

2.2.2. Dynamic time history analysis

Dynamic time history analysis (DTHA) provides a linear or nonlinear assessment of complex structural behavior under load, depending on a specific time feature. The time history of the input force or acceleration is used in time history analysis, which is then combined to produce a response. Response of the structure is calculated at multiple time instants in THA.

The Nonlinear THA permits the influence of nonlinearity of geometry and inelasticity of materials under the dynamic loading to estimate the behavior of displacement and collapse load of a structure. A basic outline of NLTHA is presented by the flow chart below in Fig. 9.

The entire time history depicts how the structure responds over time, both before and after imposed loads. The systemic response is measured at multiple time points in THA. The equation of motion of the structure needs to be computed to estimate the complete time history of the structure's response.

Lupoi et al. [25] considered RC structures & piers and established the analysis method of FF in realistic systems of structure, where it is described as simple, thorough, and computationally efficient. It incorporated DTHA to characterize the probability of demand on the structure. The ability to vary the level of sophistication in numerous ways, most notably in terms of considering or rejecting the demand's reliance on system-related uncertainty. On the contrary, when FA was performed on wood frame buildings, it was observed that geographical area, drift limit, and structure type decision did not affect the percentage of snow load. The study took into account both moderate snow and seismic dangers. The parameters produced could be used to provide acceptable percentages of design of the load in snow also to be applied (dead load being included) [26].

Moreover, Celik and Ellingwood [27] modeled beam-column joints to assess gravity load's fragility-designed RC frames. The standard rigid joint model was shown to be insufficient for replicating the extremely pinched responses of the performed experiments, highlighting the need of allowing for bond-slip and joint shear in Gravity Load Design (GLD) RC frame modeling. Besides, Seyedi et al. [28] developed the seismic fragility surface for the RC buildings through NTHA, in which several parameters were incorporated to characterize the ground motion intensity. These surfaces were developed to represent the probability of damage limit state. Fragility surfaces give a robust method in seismic risk assessment to increase the impacts of irregularity. Results obtained specify that some of the parameters that correlate to the dynamic characteristics of the building structure (example: SD for fragility analysis) are more suitable for the considered model of the structure without inculcating the effects of fatigue.

Another study outlines the preliminary research initiatives to develop a performance-based technique for estimating high-rise building response to wind threat [25]. The findings of their investigation indicated the suitability of the suggested method and the two-step MC algorithm's potential for application to more realistic cases. A similar study was focused on assessing fragility estimations for building structures with 9 stories, integrating four distinct control design layouts based on provided levels of hazard [29]. With the MR dampers application based on the approach of DPDB, the fragilities of seismic are significantly lower than those of the structures that are not controlled under two different levels of danger using 41 EQs. Structures with low-level performance targets have much lower seismic vulnerabilities than those with a greater performance within that of the similar seismic

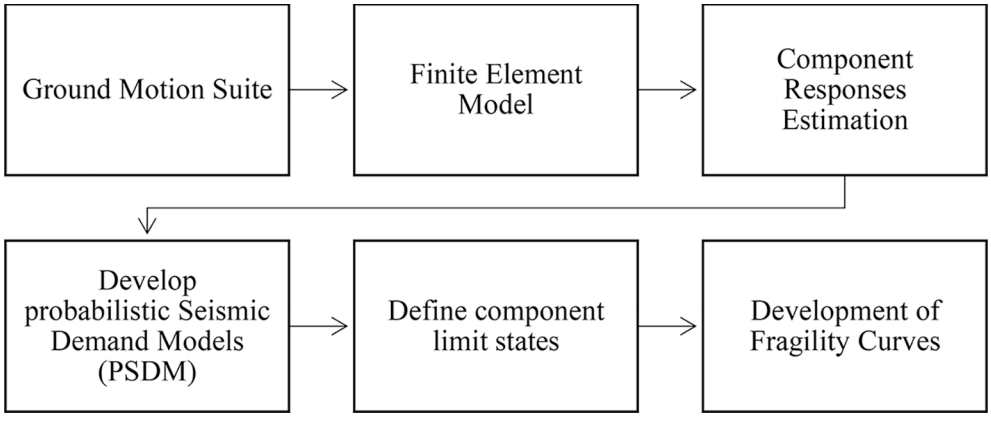


Fig. 9. NLTHA procedure for the development of fragility curves [24].

hazard category.

On the other hand, Ahmad et al. [30] derived the seismic FF for non-compliant and code-compliant RC special moment-resisting frame (SMRF) structures in Pakistan. The obtained FF indicates that non-compliant and compatible SMRFs have similar seismic properties under mild to moderate damage conditions. Still, non-compliant SMRFs outweigh the severe and critical damage conditions before compliance SMRF. Also, SMRF exhibits breakdown of joints under side loads and thereby increases the deformability of the structure due to the involvement of the confinement ties within the joint area for tensile strength and the structural strength of the concrete, which slows the. However, to demonstrate the various approaches for extracting fragility curves, a 12-story moment-resisting frame was employed. From the deterministic and uncertain EDP capacity paradigm, the resultant EDP-based results mirror those obtained using the IM-based methodology [31].

Moreover, Ghosh and Chakraborty [32] analyzed the seismic fragility based on Bayesian linear regression. Therefore, the proposed approach of Bayesian to seismic fragility analysis effectively addresses the main three aspects of fragility estimation needed for analytical seismic FA of structures, namely, computational performance, measure & efficiency of uncertainty in the estimation of fragility. Furthermore, the proposed BLR models apply to specific examples and a specific study field. Therefore, the proposed models of BLR can be used for improved seismic fragility analysis of the particular structure.

2.2.3. Incremental dynamic analysis

Incremental dynamic analysis (IDA) is a computer-assisted EQ analysis technique for determining the behavior of structures under loads of seismic. It aims to develop the results of a seismic hazard study to determine a structure's seismic risk. This can be compared to static displacement analysis in terms of dynamics. This method is advantageous, which includes running a sequence of non-linear dynamic analyses on a set of scaled ground motion records, the sensitivity of which should preferably cover the whole range of elasticity to complete dynamic instability.

Response of a Buckling Restrained Braced Frame (BRBF) is interpreted through IDA as shown in Fig. 10. Here, EDP is represented by each continuous line incorporating the influence of one ground motion record at several IM levels [33].

IDA-based analysis of fragility by He and Lu [34] suggested that by expanding the EDP limit in a particular direction by a factor of $1^2 + \lambda^2$, in which λ is the ratio of PGAs of input of records of seismic in the principal directions of two structure, one can conduct fragility analysis using combined EDPs and combined IMs while getting exceedingly probabilities approximating to results from the study. For instance, probabilities of exceedance obtained from FA indicate that tall buildings exhibit lower probabilities of exceedance when viscous dampers are installed in the structure's lower part and Bucking Restrained Braces (BRBs) are mounted in the structure's upper part. Similarly, the fragility of seismic study for structures with SRC frame using various EDPs showed that Drift-based assessment results are cautious, whereas material harm index results are risky. The results obtained for seismic risk assessments of structures with SRC frame based on the damage index of component, on the other hand, it is said to be cost-efficient, realistic, and reliable overall. Furthermore, the peak ratios-based drift inter-story & damage index of component-based fragility curves were alike for collapse prevention state fragility curve [9].

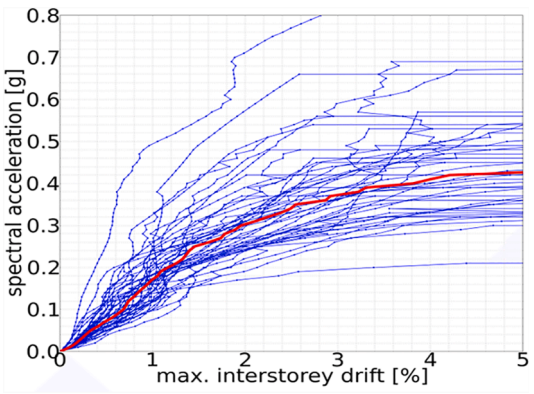


Fig. 10. Result of a typical incremental dynamic analysis [33].

Seismic performance of structures for mid-rise & normal frames using indices of combination or energy-based was more dependent on the length of records, and truncation could result in high errors. Truncating had the most negligible influence on the outcome as long as deformation-based damage indices were incorporated. The effect of truncation for the limit state of Immediate Occupancy (IO) is less than the limit state of Collapse Capacity (CP) regardless of damage index, according to the computed errors in this analysis. According to this study, truncating records could rescue more than half of the execution time [35]. Likewise, Serdar Kirçil and Polat [36] analyzed the mid-rise RC frame buildings by FA. They found that the FCs technique helps assess the damage caused by potential EQs for certain structures. The number of stories in the buildings impacts the fragility curve parameters, as seen in the FCs. It was also witnessed that a strong association between the story's number and the limit of spectral displacement at the collapse prevention performance level.

Meanwhile, Jeon et al. [37] carried out a study showing that frame vulnerability diminishes as building height rises, as the PGV median is necessary to induce a given ADS after a post-seismic events shock increases. Here, the suggested aftershock fragilities provide a better foundation for evaluating damaged structures after an EQ and establishing evacuation and re-occupancy policies. When exposed to subduction motions, Pan et al. [38] derived FCs that revealed that wood residences were more sensitive to collapse and damage. It shed light on the damage potential of pre-1970 light-frame wood buildings in British Columbia (BC). These results are also applicable to other seismically active areas as well.

On the other hand, Di Sarno and Pugliese [39] derived a probabilistic approach that is utilized to assess structure's performance under multiple excitations of a typical existing RC building and different levels of corrosion. The Modified Acceleration Spectral Intensity (MASI), a recently developed seismic intensity measure, effectively forecast the seismic fragility of corroded RC structures subjected to the mainshock-aftershock series. According to the results for all limit states, corrosion considerably impacted the seismic vulnerability of a typical existing RC building. In evaluating existing RC structures under multiple excitations and varying levels of corrosion, modern seismic standards are no longer conservative. Meanwhile, Baltzopoulos et al. [40] incorporated two case studies to illustrate the SPO2FRAG's viability as a calculating tool. The fragility function is predicted, inculcating the program that was compared and shown to be in accord with an analytical solution employing IDA. Furthermore, under the assumptions of IDA, SPO2FRAG can produce expedient answers to the related analytical issue, the fragility of seismic for the particular structure estimates for regular, symmetric frames.

2.2.4. Response history analysis (RHA)

For a newly constructed structure, validation of a new proposed design or existing structure's performance-based vulnerability assessment, non-linear RHA is carried out. The structure's response as a function of time is determined when the structural system is subjected to ground motion intensity.

Ji et al. [41] introduced an analytical framework as shown in Fig. 11 to estimate the fragility of RC tall buildings. Their prime contribution was the development of simplified lumped-parameter analytical models consisting of rigid bars, beam components, and springs that are non-linear that could be run in the ZEUS–NL setting. The simplified model

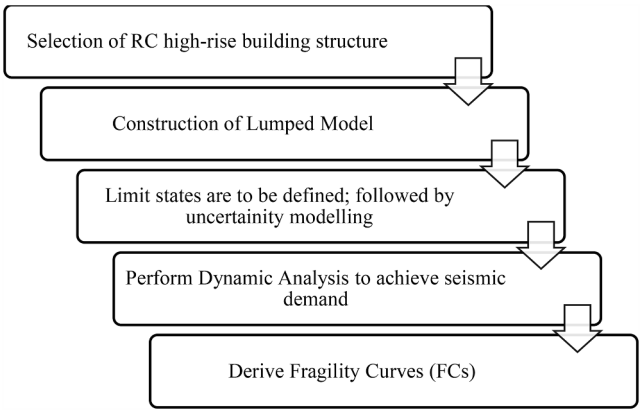


Fig. 11. An analytical framework for incremental dynamic analysis [41].

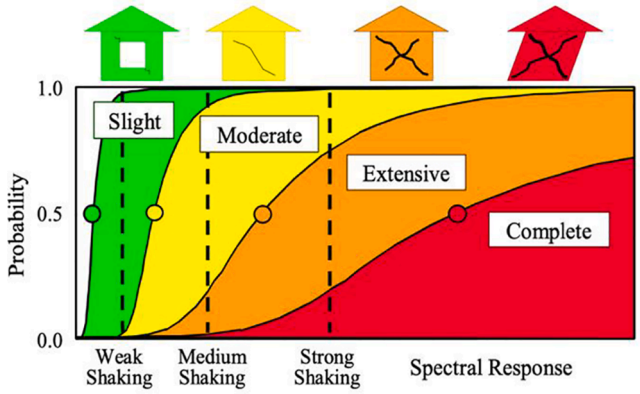


Fig. 12. Fragility curves for slight, moderate, extensive, and complete damage [5].

can represent the interactions between an external frame member and core wall over the height of the wall at each story. Also, the worked-on model's utilization brought about exceptionally critical decreases in investigation times that render it viable to create probabilistic delicacy bends for tall complex buildings, where none existed prior.

Using RHA, the dynamic responses of low & mid-rise RC structures were evaluated and showed that FCs for different numbers of stories are well-differentiated, providing insight into the expected damage distribution. Furthermore, they were effective in recreating the damage distribution observed within the 2 to 5 story buildings or structures in Düzce following the EQ in the year 1999. In functions of fragility designed for large stocks of structure or building, PGV is used as a measure of high motion strength. For large magnitude EQs, it is widely agreed that PGV strongly correlates with the modified Mercalli intensity (MMI) scale [42].

Meanwhile, Cimellaro et al. [43] investigated the influence of the ground records of scaling motion on fragility models for MDOF systems of inelastic structure.

Further, Zain et al. [44] derived analytical fragility relationships for RC buildings in seismic high-intensity zones that incorporated the effects of higher structural modes to reduce computational effort. The derived framework lessens the time to complete one calculation cycle from 72 h using traditional methods to just 4 h using a computer. The proposed methodology can be adopted to mid and low-rise RC buildings in any part of the world. Because of the more excellent modal mass participation ratios of higher modes would prove to be more effective for mid and high-rise buildings. Thus, the contribution of individual modal towards building vulnerability would possibly be evaluated more amply. Comparing the created FCs derived from dynamic RHA and the presented methodology found that the established approach for conducting vulnerability assessments is effective, quick, and simple without sacrificing accuracy.

2.3. Fragility curves

The FCs of buildings are a function of lognormal that elaborates the possibility of exceeding or reaching the damage's non-structural and structural states with the given median estimates of spectral response. The variations in probabilities of the state of damage, particularly for the primary three-stage of the response of spectrum corresponding to mild, medium, and powerful seismic ground motion, as well as examples of fragility curves for the four states of damage used in the FEMA/NIBS technique.

Fig. 12 represents the examples of FCs that express the varying probability of damage state, i.e., slight damage (SD), moderate damage (MD), extensive damage (ED), and complete damage (CD) corresponding to three spectral response levels [5].

Mai et al. [45] presented methods that can be used to generate FCs when their exact shape is unknown and also to calibrate and validate parametric fragility curves. It was also discovered that the FCs derived by explicitly addressing the ground motion intensity uncertainty were flatter than those generated without the uncertainty [46]. Moreover, the FCs of Surana et al. [47] reveal that the slope angle significantly impacts seismic efficiency.

In contrast, Kiani et al. [48] identified how machine learning approaches might be used to anticipate the responses of the structure and derive FCs. Their general approach proposed can be utilized to predict the parameters and responses of different structural systems' FCs.

3. Past studies on fragility analysis of structures

3.1. Class of buildings

According to their structural characteristics, researchers have categorized several groups of buildings to develop the fragility models to analyze the response or behavior of structure for general building class and estimate the expected damage. Table 3 represents the details of building classes adopted in this study.

In the analysis performed by Liao et al. [49], the FCs of various types of buildings were formed. The findings can be incorporated to improve the Taiwan EQ Loss Estimation System (TELES) module for the general building system and estimate seismic losses. A few generalizations were proposed for computation clarity because the data collected is limited. However, Yazgan [46] suggested fragility models directly integrated into risk management systems and current seismic loss estimation frameworks. It was discovered that the FCs derived by explicitly addressing the ground motion intensity uncertainty were flatter than those generated without uncertainty.

The framework proposed by Gernay et al. [50] includes the various sources of uncertainty that affect the steel building's fire vulnerability, including the scales of building and material. The developed functions of fragility can be inculcated to assess the likelihood of a fire disaster in a group of buildings. Meanwhile, Villar-Vega et al. [51] addressed the production of a common model for fragility for the fifty-four classes of a building of the Andean countries. Depending on the building class, varying ductility levels, response behaviors and damage schemes were considered. A cumulative lognormal distribution was used to model the fragility function. Based on the association between damage distribution and shaking of the ground; the representative intensity measure level was chosen.

On the contrary, Kohrangi et al. [52] showed that when average spectral acceleration (AvgSA) is employed instead of SAT1, the spread in the identical buildings' response placed at various sites is considerably reduced. Using AvgSA minimizes the impact of varied ground motion characteristics was expected on the projected response significantly in

Table 3
Description of adopted building classes in the referred literature

Author(s)	Building class	Typology	Material type	Adopted methodology
Liao et al. (2006)	15 classes of buildings in TELES	• High-rise (8+ stories) • Mid-rise (4-7 stories) • Low-rise (1-3 stories)	• Steel frame • Concrete frame • Masonry walls • Steel Reinforced	HAZUS
Yazgan (2015)	3 classes of RC buildings	• RC 12 • RC 34 • RC 56	• RC moment-resisting frame (MRF)	Proposed a new framework under empirical approach
Gernay et al. (2016)	Entire Steel building	• Steel frame building (9 stories)	• Steel frame building	To develop fire fragility, a new methodology was proposed
Villar-Vega et al. (2017)	54 classes of buildings located in Andean Countries	• IF (1-7 stories) • RC MRF (1-7 stories) • RC dual frame (10 stories) • RC wall system (5-10 stories) • CM (2-3 stories) • URM (1-3 stories) • Lightwood (1-2 stories)	• Brick masonry • RC • Earthen or Adobe • Wooden • Stone Masonry • Non-Engineered	Referred from guidelines supported by Global Earthquake Model (GEM)
Kohrangi et al. (2017)	Class of Steel frame and RC frame buildings	• Steel frame (4 stories) • 3 RC frame (7,12 & 20 storeys)	• RC frame • Steel frame	Considering the effect of multiple sites, they provided two methods
Del Gaudio et al. (2020)	Classes of existing RC buildings in Italy	• Rc buildings	• Masonry • RC • Others	A new methodology was proposed to derive fragility curves.

many varying sites. As a result, even if record selection is not applied, AvgSA can be utilized to mitigate the unfavorable effects of this omission. AvgSA's alternate benefits, such as its efficiency in predicting the local, including the global responses, and its ability to serve as a single common IM for a building class, make it an ideal but realistic alternative for risk assessment and portfolio loss.

Within the FA and damage due to seismic, the relevance of treatment of damaged data and accurate preliminary analysis has been presented and explored by Del Gaudio et al. [53]. In addition, the predictive efficiency of the produced curves was discussed, emphasizing the necessity of using no-damage data for low-intensity fragility research.

3.2. Unreinforced masonry (URM) buildings

Under-developed and developing countries were the top places with unreinforced masonry (URM) structures that are non-engineered and have unsafe construction practices. The vulnerability of such structures is at its peak, i.e., they are prone to damages majorly in seismic excitation. Therefore, a low-intensity EQ event could easily lead to the enormous destruction of such structures, followed by many casualties. FCs were developed for the analytical assessment based on the model for a masonry structure with 2 stories that are unreinforced, which is considered typical in the central and southern US (CSUS). The seismic performance for URM structures is considerably underneath the level of performance for seismic for the desired building indicated by the seismic code, revealing the severe URM building's vulnerability, according to the FA [54].

Moreover, the seismic risk of the URM building typology in Northeast Indian cities could be assessed using the FCs derived by Halder et al. [55]. Due to poor DCQ, buildings were observed with huge damages even with a seismic event's most minor PGA level of 0.052 g. Fig. 13 shows developed cracks in a single-story URM building incorporating a lightweight galvanized iron (GI) sheet. The obtained fragility curves indicate that the seismic vulnerability of residential URM buildings located in Agartala is relatively high. The generated FCs could be utilized to determine the seismic risk of other regions, except that it applies to structures of the same type.

Besides, Battaglia et al. [56] performed a case study where the principal objective was to see if there were any changes in the seismic behavior of the buildings of masonry aggregate when evaluated as isolated structural components or when analyzed a row aggregate of similar structural elements. It showed how changes in the chosen structural and geometrical characteristics substantially impact the seismic response. First, internal wall type was discovered to be a significant fragility factor, followed by the most critical factor influencing seismic response was the type of slabs (either timber or concrete). Additionally, the torsional impacts are lessened in the timber slabs, whereas torsional effects occur when concrete slabs are present.

On the contrary, Miano et al. [57] suggested methods that provide a realistic way to account for uncertainties in the anticipated ground shaking level and ground motion correlations at the damaged structure's locations. The statistical update of recorded ground motion is critical in adequately combining local site variables and ground motion levels. The suggested method can combine damage estimation methods employing remote sensing techniques for faster fragility estimation.

The fragility model based on typical URM brick-in-cement buildings in Malawi is shown in Fig. 14. It combines structural material research with field data on building geometry. The main focus was that non-engineered Malawian houses are highly susceptible to out-of-plane damage, which might be avoided by improving construction features like connections of wall-to-roof and wall-to-wall. Finally, the derived FCs constitute a significant step forward in executing seismic risk modeling studies in Malawi and the larger East African region [58].

Meanwhile, Giordano et al. [59] compare statistics from various geographical areas and sources to produce an overview of the available data within the Nepalese school building. It emphasizes the need for more experimental studies on traditional Nepali masonry to disprove some of the model's assumptions. The findings can be easily modified as more country-specific data become accessible using the same approach.

3.3. Precast structures

Precast (PC) structures are known for their quality and rapid construction, minimal maintenance, higher serviceability, etc. However, despite its various advantages, PC structures also faced tremendous damage in past seismic excitation. Therefore, regarding this study, fragility analysis on PC structures was found lesser than typical RC structures.

A set of fragility curves derived by Casotto et al. [60] took into account the critical characteristics of the structure, namely the connection of the weak beam and column, that are the primary basis of a complete collapse of building due to lack of beam support, even in milder EQs. The assessment of the FCs for collapse with and without connection failure revealed the need to consider and describe the connection collapse mechanism. Whereas, Babič and Dolšek [61] investigated precast buildings that are single-storeyed. Based on their study, in the design of buildings, components of non-structure should not be ignored, nor when evaluating their seismic efficiency or risk. The utilization of masonry infills in buildings with a Type 2 structural configuration and low-code design, the median PGA is causing huge collapse reduction by 35%. A specific building's FF for a single seismic event cannot be incorporated to evaluate the damage.

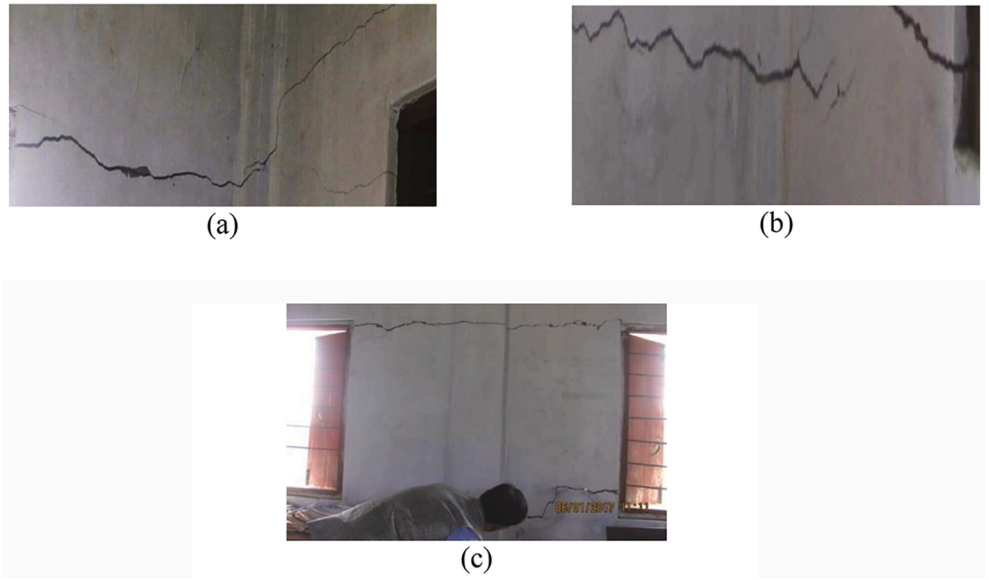


Fig. 13. (a,b) The orthogonal wall at corner joints found with large shear cracks, (c) Wall damages observed in between the openings of a window [55].



Fig. 14. Typical Malawian single-story brick-in cement masonry building. (a) A light-weight corrugated metallic roof supported by lightweight timber truss, (b) Typical flat roof) [58].

3.4. Soil-Structure interaction

The numerical parametric study considered various idealized slope configurations, soil and geological settings, structure distances from the slope's crest, and foundation typologies. In conjunction with the slope soil material, the degree of inclination of the slope has a remarkable influence on the building's vulnerability to differential permanent deformation caused by a seismically induced landslide hazard [62]. When Peduto et al. [11] focused on 49 buildings that are exposed, including 37 of structure that is masonry with shallow foundation and 12 RC buildings, also covered by at least two Differential Interferometric Synthetic Aperture Radar (DInSAR) benchmarks in both COSMO-SkyMed and ENVISAT datasets (phase I). The findings reveal a general upward tendency in the above severity levels with low-intensity landslides, regardless of the scope of the investigation or the structural typology of the buildings under consideration. Furthermore, the preliminary creation of the FC and empirical fragility, including both the aggregates of the masonry building in areas that are vulnerable (at a medium scale) and masonry buildings structures that are single, was made possible by the use of such widely available data as DInSAR data (at large scale).

A consistent approach was developed to estimate the structure's vulnerability near the crest of precarious landslide slopes in a post-seismic event. The developed FCs could be utilized to predict the physical vulnerability of RC buildings that are low-rise exposed to displacements of slope due to EQs. The analytical equations for co-seismic slope displacements (PGD or D) developed could be employed as IM [63].

Moreover, Su et al. [64] assessed seismic fragility on large-scale pile-supported wharf structures (PSWS) considering soil-pile interaction (SPI). The SPI has an effect that is notable on the PSWS over-analysis. The difference between the heap ejection response with and without SPI was found to be very significant for F. The contrast between the required FC without & with SPI becomes more pronounced from damage states of slight to extensive, revealing the modelers that the effect of SPI should be more considered when evaluating the fragility of seismic of the PSWS under extended hazard status.

Meanwhile, Zhai et al. [10] main focus were to equate the effects of SSI on the fragility of structure to those of the fixed-base model. The findings indicate that using both the double-parameter damage models (DPDMs) and the IMDR to account for SSI effects causes the structure to be more fragile than the fixed-base model, making the structure unsafe. As a result, the impact of SSI should not be ignored when assessing the fragility of structure. Moreover, in part to escape conservative bias in the assessment of risk outcome, it might find the necessity to not only incorporate effects of SSI in the implementation of the FF of stock of building but rather to use a non-linear model of SSI; particularly for buildings built on the piles and hence impacted by the foundation system's rocking response [65]. Furthermore, the fragility models proposed by Karafagka et al. [66] can be inculcated for the vulnerability to assess

and for the risk of RC frame buildings that are non-ductile in Southern Europe that are prone to liquefaction and ground shaking. It was discovered that the extreme soil's non-linear behavior caused by liquefaction of the soil causes the motion of the seismic event at the surface of the ground to be attenuated not in terms of velocity but acceleration.

3.5. Buildings sited in hilly region

Structures in uneven ground level or hilly regions always witness much higher damage during a particular seismic excitation. Therefore, the authors addressed fragility analysis to such structures to understand the post-EQ structural phenomena.

Sreerama and Ramancharla [67] showed from their obtained FCs that the structure is much more vulnerable when it is at a steep angle. As the slope angle tends to increase, a hinge mechanism forms near the shorter column zone and is damaged sooner. For buildings sited in the region of slopes, the maximum inter-story drift of a three-story, five-span building on the hillside is susceptible at the first two performance levels but insensitive at the last two, which can be explained by the fact that the interforce is redistributed among columns. The fragility analysis revealed that the architecture of buildings on the hill has a significant impact on the efficiency of seismic [68]. Similarly, Huang et al. [69] also mentioned that slope has a considerable effect on seismic efficiency; hence, a control method to decide the best configuration could be established further.

On the other hand, Surana et al. [47] clarified that the existing seismic design code (India), including other countries, is inadequate for building structures on the hill slopes since they were designed for buildings on flat ground. Additionally, minor changes in structural configurations that result in a more normal distribution of strength and stiffness can improve performance. Moreover, hill building layouts are more vulnerable to destruction than ordinary buildings on flat land. Surana et al. [70] compared high-code and pre-code buildings. The impact of the configuration of the particular structure on the fragility of seismic was found to be more evident in pre-code buildings. Pre-code hill buildings have a 50% lower median collapse capacity than their flat-terrain counterparts, and high-code buildisngs have a 10% lower median collapse capacity. The failure of shear of the short columns and irregularity of torsion contribute to the increased fragility of hill structures.

Furthermore, Surana et al. [71] studied various RC frame structures (Without infill walls) configurations of FL, SF, and SB buildings in hilly regions, as shown in Fig. 15. The story right above the uppermost foundation level was the most susceptible to vulnerability in SB and SF buildings. Comparing the collapse probability of buildings inculcating various structure configurations shows that the existing code requirements are appropriate in the case of low and mid-rise buildings in the zone of Seismic IV (moderate magnitude). Still, they were insufficient for tall buildings for the zone of seismic IV and buildings in the Zone of seismic V. Also, the collapse fragilities could result in significant changes if the study can be broadened by considering the infill walls. The research sheds light on the seismic behavior of irregular structures in the hillside. It emphasizes the requirement for further research into design codes that consider the configuration of the structure, level of seismicity, height of building, and impact of the near field.

3.6. Plan irregularity

The irregularities of the structure considerably influence the Probabilistic Seismic Demand Model (PSDM) parameters. It was also observed that the construction quality (CQ) and soft-story (SS) significantly impact seismic vulnerability. As a result, in risk assessment of seismic, consideration of the quality of construction and vertical irregularities has a considerable impact on the phase of decision-making [72]. Moreover, Moon et al. [73] proposed a method for calculating the fragility curves of seismic allows for the use of computationally expensive models in the analysis while also producing very efficient fragility curves. It calculates the functional curves of seismic fragility for the particular space RC frame structures inculcating varying plan irregularities. Also, it uses more realistic fragility curves to look into the impact of plan irregularity on seismic vulnerability.

Further, when Bhasker and Menon [74] contrasted to those based on structure-specific measures, it showed that vector IMs based on non-structure specific IMs are less effective in estimating peak inter-story drift demands. The study emphasized the heteroscedastic nature of all three IMs, emphasizing the importance of explicitly taking this into account when using a cloud analysis-based demand model to derive fragility functions. The reduced inter-record dispersions in demand predicted by the optimal intensity measure led to more reliable estimates of seismic fragility.

3.7. Retrofitted structures

Seismic FA was carried out on an RC flat-slab building that has 5 stories with the typical type of construction in the 1980 s in the region of Mid-America. The seismic FCs developed in this work helped calculate the existing structure's vulnerability in the United States. The FEMA 356 level of member-based drift limits tends to be much lower than the FEMA 356 limitations of global level [75]. Shafaei and Naderpour [76] mentioned that while the retrofit strategy is beneficial in raising the structure's collapse capacities, the capacity reductions induced by various mainshock damage levels in the analyzed structures are independent of the structure's ductility and strength capacities, according to the findings. The capacity of collapse for ordinary reinforced concrete frames (ORCFs) retrofitted with sheets of Fibre Reinforced Polymers (FRPs) and wrapped under mainshock-aftershock sequences. Regardless of the retrofit's height and presence or absence, the median collapse capacities (MCCs) obtained for various damage levels revealed somewhat similar trends. However, due to the introduction of damage of the mainshock, the trend showed a sudden initial reduction in aftershock MCC.

Moreover, Hueste and Bai [71] carried out a Seismic fragility analysis on an RC flat-slab building with 5 stories with the typical type of construction in the 1980 s in the region of Mid-America. The seismic

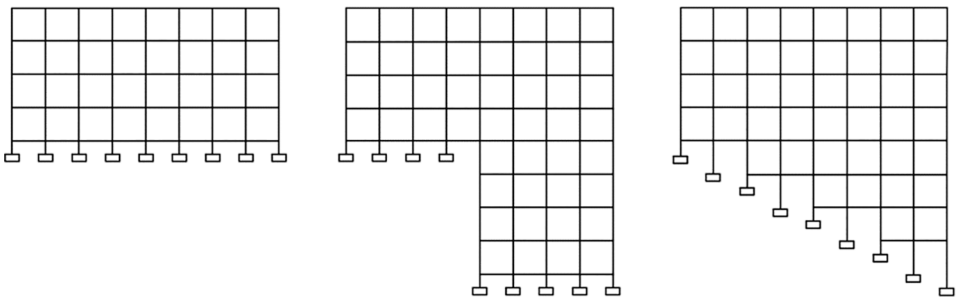


Fig. 15. (a) Flat land (FL), (b) Split-foundation (SF) building, and (c) Step-back (SB) building [71].

fragility curves developed in this work helped calculate the existing structure's vulnerability in the United States. The FEMA 356 level of member-based drift limits tend to be much lower than the FEMA 356 limitations of the global level.

3.8. Transmission towers

A case study was conducted by Pan et al. [77] on a Transmission Tower (TT). The seismic response and seismic fragility were found to be influenced far more by uncertainties in material properties than by uncertainty in geometric dimensions. There is a large discrepancy between the deterministic and uncertain model fragilities, demonstrating that the structural uncertain factors have a decent impact on TT fragilities. The employed TT was adequately developed aiming to prevent collapse for minor, basic, and major earthquakes; nonetheless, when all structural uncertainties are taken into account, the tower may collapse under a severe earthquake. However, Tian et al. [78] incorporated the probabilistic seismic demand model (PSDM) created in terms of the maximum inter-segment drift ratio (ISDR) and spectral acceleration (Sa) during the fundamental period of the structure to study the seismic fragility of TT. When compared to the linear demand model, the bilinear demand model was more precise and reasonable. The seismic incidence angles considerably impact the transmission tower's fragility. Also, the seismic fragility of the transmission tower might be exacerbated by the dynamic coupling effect between the TT and lines. Without the coupling effect, it might lead to an overestimation of the TT's seismic capacity.

3.9. Seismic system reliability of complex slopes

The system reliability of complicated slopes is investigated using the recently published Generalized Probability Density Evolution Method (GPDEM), which takes into account uncertainty in numerous slope parameters as well as ground motions [79]. While maintaining adequate precision, GPDEM was found to be 10–100 times more efficient when compared to Monte Carlo Method (MCM). The findings show that the proposed GPDEM is capable of evaluating nonlinear geotechnical dynamic situations with uncertainty. It also observed that randomness in ground motions has a considerably higher impact than randomness in slope parameters, yet slope parameter's randomness cannot be fully ignored because it could lead to substantial changes in the failure probability evolution.

3.10. Concrete face rockfill dam (CFRD)

The influence of aftershocks on the fragility of high concrete face rockfill dam (CFRD) is investigated using a 200 m high CFRD as an example [80]. The most relevant and effective EDP for studying the cumulative damage produced by aftershocks to the high CFRD is vertical deformation. After the aftershocks, the CFRD's horizontal deformations may intensify or decrease. High CFRD based on vertical deformation is more vulnerable to aftershocks than those based on Damage Index (DI). Under the same combination of mainshock aftershock sequences, the dam damage risk based on DI is larger than that based on vertical deformation.

Moreover, fragility analysis was used to conduct a seismic safety evaluation on a high CFRD [81]. The probabilities of each seismic performance assessment indicator were calculated for various earthquake actions, and a standard or reference for dam seismic performance under major earthquake was established. Their study examines the dam's failure probabilities under various earthquake intensities and can serve as a reference and basis for high CFRD seismic performance design and risk assessment.

Another research work by Pang et al. [82] presents a method for analysing the seismic performance of a CFRD based on deformation and damage index and proposed a seismic fragility analysis based on IDA to the seismic safety assessment area of high CFRDs. The fragility technique, which took into account the randomness of the ground motion inputs, gave a thorough assessment of the dam's seismic performance and a reference for the dam's seismic performance during a large earthquake. The relative settlement ratio of the dam crest and the damage index were used to provide standards for differentiating dam failure grades, and three failure levels of seismic performance were determined: minor, moderate, and severe failure.

Additionally, the accuracy and efficiency of a recently developed generalised probability density evolution method (GPDEM) coupled with a spectral representation-random function method were compared to those of the Monte Carlo method (MCM) for strongly nonlinear structures in high CFRDs during earthquakes [83]. It was observed that the GPDEM is substantially more efficient (almost 10 times faster) than the MCM while maintaining sufficient precision. As a result, the GPDEM provides a realistic, cost-effective, and efficient analytical tool for stochastic seismic investigation of high CFRDs.

4. Discussion

This study summarises the fragility analysis carried out on various structures by several researchers.

- Huge Structural EQ damages were observed in India in the past. The population is one major factor for the increasing scarcity of land in urban areas. Construction of open ground story has been a common practice to allot the area for usage as parking or commercials. This is one of the prime reason for extensive damages in a structure due to seismic excitation.
- Seismically highly active regions of India, such as the Northeast region and other developing countries around the globe such as Pakistan, Nepal, etc., have witnessed substantial structural damages during the past EQs. The major reasons were the poor design construction quality, and lack of maintenance during the construction.
- It can also be highlighted that the development of fragility curves for confined masonry construction is least explored domain among the researchers.
- In seismic fragility analysis, soil pile interaction (SPI) might negatively or positively affect various structural components and various damage states. Moreover, the effect of soil-structure interaction (SSI) should not be ignored. The overall height of a building has substantial impact on its susceptibility to liquefaction of soil and shaking of the ground.
- It was found that the height of a slope has considerable impact on the fragility of a building, in case the layout is steep sand slopy.
- Considering the region of hills, existing seismic design codes were found deficient for high-rise buildings.

5. Future scope

The following areas could be explored in the future.

- Design of reinforced concrete special moment rrsisting frame structures could be done considering several vulnurable conditions, such as variation of resting angle with the groundto serve as a future reference by predicting its possible damage probabilities.
- The impact of structural uncertainty on transmission tower coule be explored through the fragility curves.
- The structure facing mainshock-aftershock sequences of high intensity earthquakes could be studied by linking various intensity measurements and engineering demand parameters on high concrete filled rockfill dams.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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